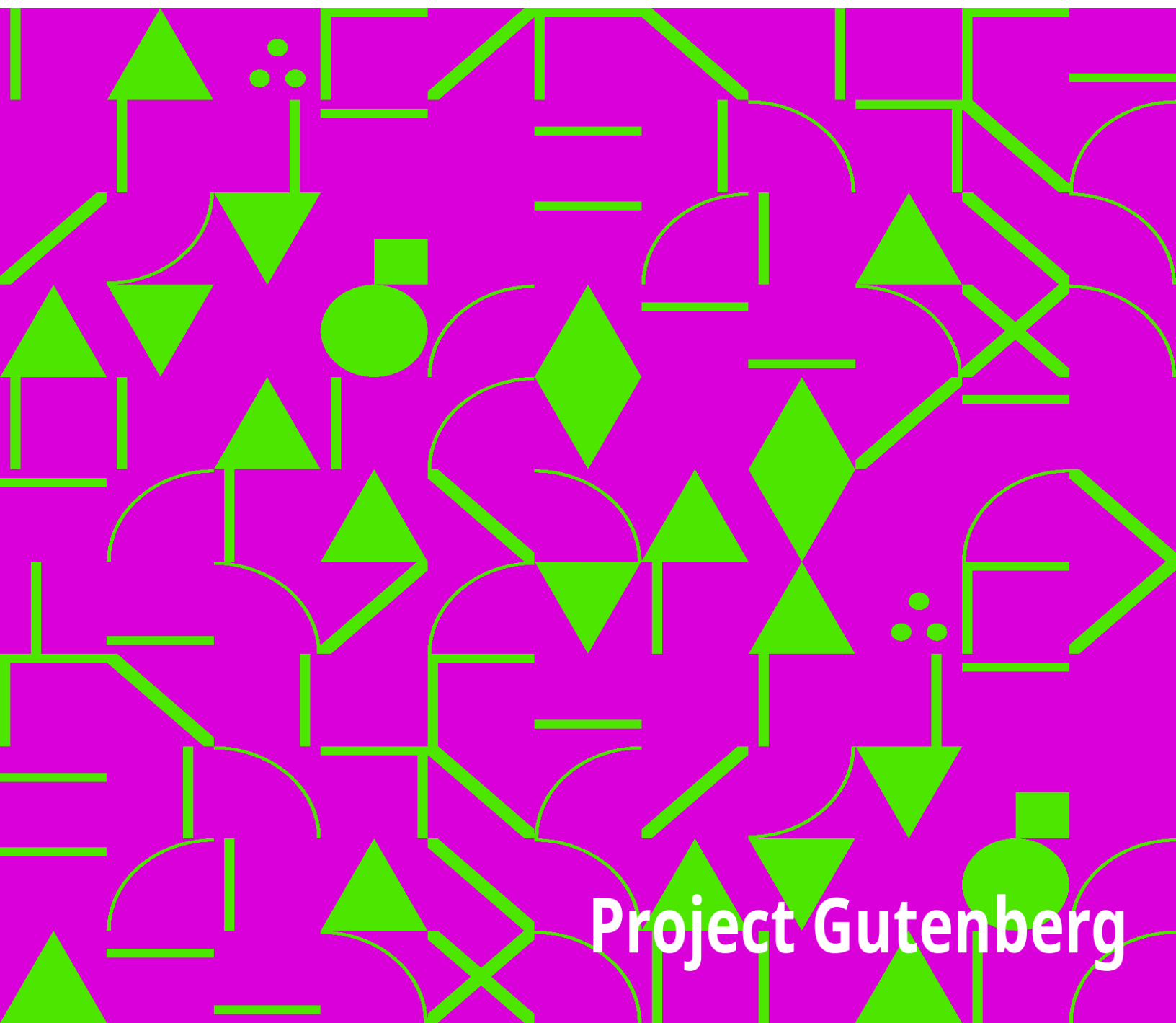


Taxonomy of the Chipmunks, *Eutamias quadrivittatus* and *Eutamias umbrinus*

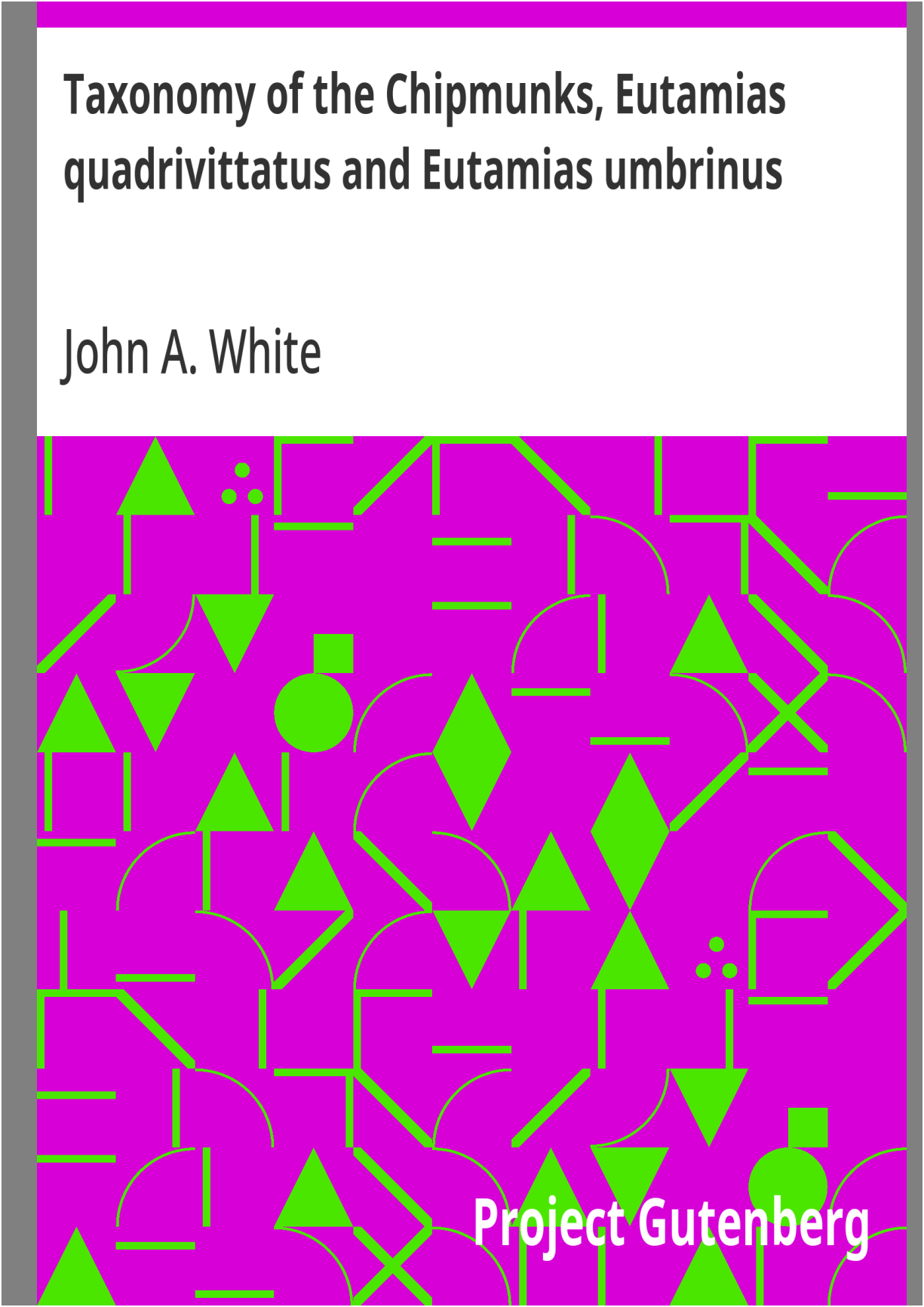
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Title: Taxonomy of the Chipmunks, Eutamias quadrivittatus and Eutamias umbrinus

Author: John A. White

Release date: February 12, 2010 [eBook #31267]

Most recently updated: January 6, 2021

Language: English

Other information and formats: www.gutenberg.org/ebooks/31267

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**University of Kansas Publications
Museum of Natural History**

**Volume 5, No. 33, pp. 563-582, 6 figures in text
December 1, 1953**

**University of Kansas
LAWRENCE
1953**

UNIVERSITY OF KANSAS PUBLICATIONS, MUSEUM OF NATURAL HISTORY

**Editors: E. Raymond Hall, Chairman, A. Byron Leonard,
and Robert W. Wilson**

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UNIVERSITY OF KANSAS
Lawrence, Kansas

PRINTED BY
FERD VOILAND, JR., STATE PRINTER
TOPEKA, KANSAS
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24-8966

Taxonomy of the Chipmunks, *Eutamias quadrivittatus* and *Eutamias umbrinus*

By

JOHN A. WHITE

The differences in anatomy and color between many species of chipmunks are subtle, and refined techniques are required to discover them. When "measuring" chipmunks taxonomically, it is necessary to use a "chipmunk scale" and not, for example, a "pocket-gopher scale." In explanation, some species of pocket gophers closely allied to each other, and even some subspecies of the same species, differ markedly in color and in size and shape of parts of the skeleton; comparable differences are not so pronounced among many species of chipmunks.

HISTORICAL SUMMARY

Merriam (1905) was the first to show clearly that *Eutamias quadrivittatus* is a distinct species, and pointed out that *E. amoenus operarius* (= *E. minimus operarius*) is a small species which resembles, and is found in some areas together with, *E. quadrivittatus*.

Howell (1929) placed under *E. quadrivittatus* the following subspecies: *E. q. quadrivittatus*, *E. q. hopiensis*, *E. q. inyoensis*, *E. q. frater*, *E. q. sequoiensis*, and *E. q. speciosus*.

Hardy (1945) placed *E. adsitus* under *E. quadrivittatus* as *E. q. adsitus*, and Kelson (1951) placed *E. umbrinus* under *E. quadrivittatus* as *E. q. umbrinus*.

Johnson (1943) re-established *E. speciosus* as a separate species, and in California left only *E. q. inyoensis* in *E. quadrivittatus*.

Thus, since 1943 the recognized subspecies of *E. quadrivittatus* have been: *E. q. quadrivittatus*, *E. q. hopiensis*, *E. q. inyoensis*, *E. q. nevadensis*, *E. q. umbrinus*, and *E. q. adsitus*.

METHODS, MATERIALS, AND ACKNOWLEDGMENTS

Capitalized color terms, which are used in descriptions and comparisons, are of Ridgway, "Color Standards and Color Nomenclature," Washington, D. C., 1912.

In the synonymy of each subspecies there appears only the first usage of a name, second the first usage of the name combination now employed unless a new combination is proposed by me, and third pure synonyms. The last is recognizable as such because the type locality is appended to each.

Unless otherwise specified, all specimens are in the Museum of Natural History, University of Kansas. The various collections of institutions and of private persons are indicated by the following symbols:

- AM—American Museum of Natural History.
- BS—United States Biological Surveys Collection.
- CM—Colorado Museum of Natural History.
- DC—Collection of Donald R. Dickey (now the collection of the University of California at Los Angeles).
- FC—Collection of James S. Findley.
- KU—Museum of Natural History, University of Kansas.
- MM—Museum of Zoology, University of Michigan.
- NM—United States National Museum.
- UU—Museum of Zoology, University of Utah.
- WC—Collection of Edward R. Warren, Colorado College.

Of the external measurements, only the total length and the length of the tail are recorded in table 1. Some field collectors measured the ear from the notch and others from the crown; most collectors measured the length of the hind foot to the nearest millimeter rather than in tenths of a millimeter, as would have been desired. Consequently, I decided against using the lengths of the ear and hind foot in the study here reported on.

The measurements of the skull were made as shown in figure 1.

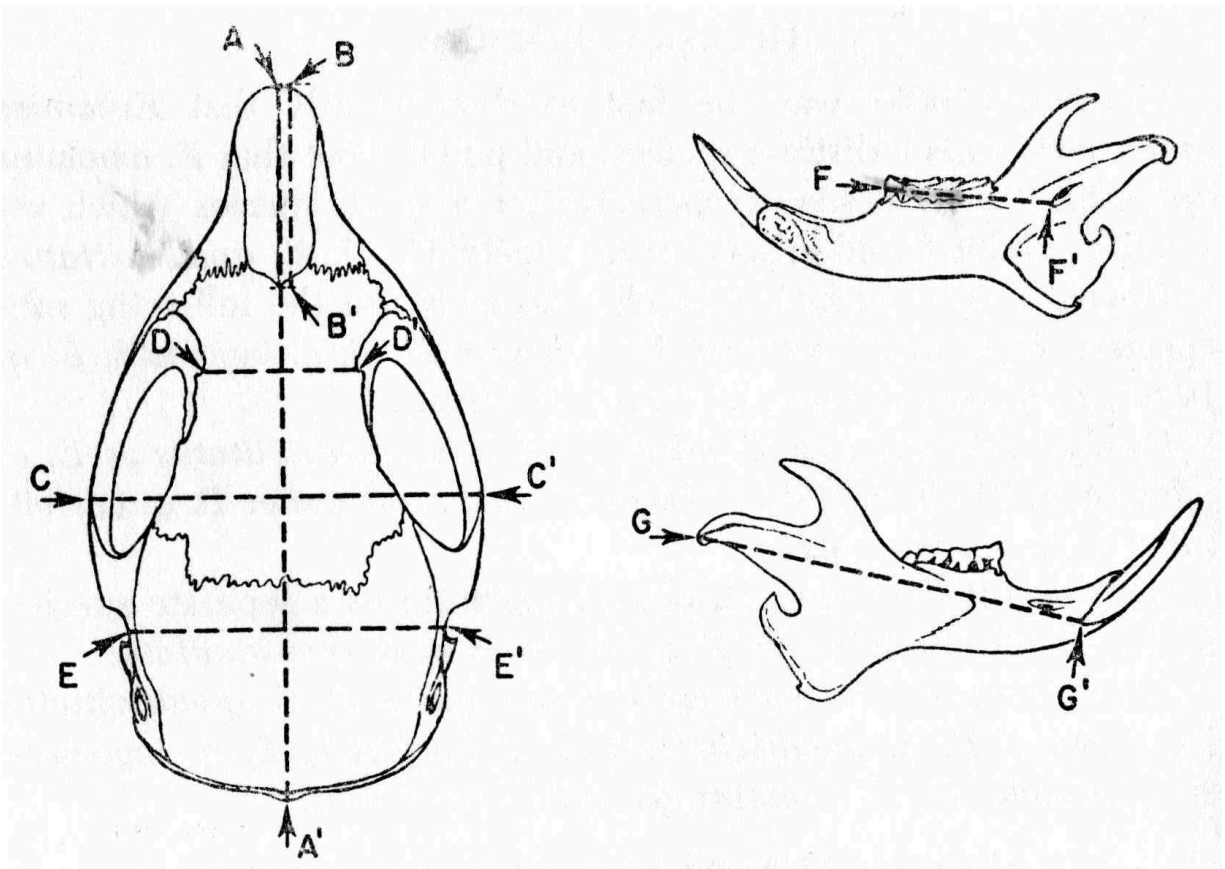


FIG. 1. Dorsal view of skull and a lateral and a medial view of the right lower jaw to show points between which measurements of the skull were taken. $\times 1\frac{3}{4}$. Based on *Eutamias ruficaudus ruficaudus*, from 6 mi. S St. Mary, 6500 ft., Glacier Co., Montana. A to A'—greatest length of skull; B to B'—length of nasals; C to C'—zygomatic breadth; D to D'—least interorbital constriction; E to E'—cranial breadth; F to F'—inner mandibular length; G to G'—condylo-alveolar length of mandible.

A total number of 434 specimens are listed as examined in this study, and additionally, numerous other specimens were superficially examined in the United States Biological Surveys Collection. Bacula of each of the named kinds of chipmunks in this paper, were examined.

Whenever two or more samples are stated to be significantly different, the meaning is that the difference is statistically significant.

The geographic distribution of each subspecies and the localities of specimens or series of specimens are plotted on the map (fig. 2).

When comparisons were made to ascertain specific and subspecific differences, only adults, or animals in which the enamel was worn through on the permanent P4 and p4 were used. Within this age range, only specimens in comparable pelage were used to ascertain differences in color.

Miss Viola S. Schantz of the United States Fish and Wildlife Service, Mr. Alfred Bailey of the Colorado Museum of Natural History, Dr. W. H. Burt of the Museum of Zoology of the University of Michigan, Dr. Stephen D. Durrant of the Museum of Zoology of the University of Utah, Dr. Robert M. Stabler, curator of the Warren Collection of Colorado College, and Mr. James S. Findley, generously loaned specimens for my use. Doctors E. Raymond Hall, Rollin H. Baker, Robert W. Wilson, Keith R. Kelson, E. Lendell Cockrum, and other friends and associates have given valued suggestions and assistance. My wife, Alice M. White, made the illustrations and helped me record and analyze the data.

Assistance with field work is acknowledged from the Kansas University Endowment Association, the National Science Foundation, and the United States Navy, Office of Naval Research, through contract No. NR161 791.

ACCOUNTS OF SPECIES AND SUBSPECIES

***Eutamias quadrivittatus* (Say)**

Diagnosis.—Size medium; general tone of upper parts tawny; cranial breadth averaging between 16.0 and 16.8 mm.; baculum distinguishable from that of any other species by the combination of width of base less than $\frac{1}{4}$ of length of shaft, shaft having a maximum diameter of more than $\frac{1}{4}$ mm., and height of keel $\frac{1}{4}$ of length of tip.

***Eutamias quadrivittatus quadrivittatus* (Say)**

Sciurus quadrivittatus Say, in Jones, Long's Expedition to Rocky Mountains, 2:45, 1823.

Eutamias quadrivittatus, Miller and Rehn, Proc. Boston Soc. Nat. Hist., 30:43, December 27, 1901.

Tamias quadrivittatus gracilis J. A. Allen, Bull. Amer. Mus. Nat. Hist., 3:99, June 1890, Type from San Pedro, Santa Fe Co., New Mexico.

Eutamias quadrivittatus animosus Warren, Proc. Biol. Soc. Washington, 22:105, June 25, 1909. Type from Irwin Ranch, Las Animas County, Colorado.

Type.—None designated; from along Arkansas River, about 26 mi. below Canon City, Fremont County, Colorado; obtained on July 18, 1820.

Diagnosis.—Size medium; dorsal dark stripes blackish; sides Cinnamon to Clay Color; crown Light Drab; baculum large.

Description.—*Color pattern*: Head Cinnamon, shaded on crown to Light Drab; ocular stripe Fuscous Black, with Cinnamon along margins; other facial stripes Fuscous mixed with Cinnamon; ears Fuscous Black, Ochraceous-Tawny on anterior margin, grayish white on posterior margin and on postauricular patch; dark dorsal stripes black with Ochraceous-Tawny along margins; outer pair of dark stripes often mainly Tawny; light dorsal stripes grayish white, outer pair usually creamy white; sides Ochraceous-Tawny, shaded in the region of the shoulder with Cinnamon; rump and thighs

Cinnamon-Buff mixed with Smoke Gray; antipalmar surfaces of forefeet Cinnamon-Buff; antiplantar surfaces of hind feet Pinkish Buff; dorsal surface of tail Fuscous Black, overlaid with Pinkish Buff; ventral surface of tail Ochraceous-Tawny, Fuscous Black along margin, Pinkish Buff along outermost edge; underparts creamy white. *Skull*: Large; braincase well inflated; zygomatic arches strong and slightly appressed to skull. *Baculum*: Large; long and slender.

Comparisons.—From *E. q. hopiensis*, the only other subspecies in this species, *E. q. quadrivittatus* differs in: Dorsal dark stripes blackish; crown grayer; rump and thighs grayer; general tone of upper parts darker.

Remarks.—Specimens from the Chuska Mountains, Zuni Mountains, and Blanco, New Mexico, are intergrades between *E. q. quadrivittatus* and *E. q. hopiensis*, but are referable to *E. q. quadrivittatus*.

In north-central Colorado *E. umbrinus* occurs in the spruce and pine forests at higher altitudes, while to the south and east of this area *E. q. quadrivittatus* occurs in growths of piñon in lower, semiarid areas. In the northern half of New Mexico and in south-central Colorado, *E. q. quadrivittatus* occurs not only in semiarid habitats but also in the moist habitats of the forests of higher altitudes. Ecologically, *E. umbrinus* thus replaces *E. q. quadrivittatus* in north-central Colorado. This ecological replacement is comparable to the ecological replacement of *Thomomys bottae* by *T. talpoides* in Utah as shown by Durrant (1952:156).

Specimens examined.—Total number, 130.

Colorado: *Larimer Co.*: Arkins, 1 BS. *Jefferson Co.*: W spur Lookout Mountain, near Golden, 1 WC. *Gunnison Co.*: Sapinero, 3 BS. *Saguache Co.*: 5 mi. N and 22 mi. W Saguache 10,000 ft., 1; 21 mi. W and 3 mi. N Saguache, 1. *Fremont Co.*: 18 mi. S and 7 mi. W Colorado Springs, 1; Arkansas River, "about" 26 mi. below Canon City, 15 BS. *San Juan Co.*: Silverton, 1 BS. *Mineral Co.*: 3 mi. E Creede, 1. *Alamosa Co.*: Sangre de Cristo Range, 24 mi. E Hooper, 2 CM. *La Plata Co.*: 2 mi. NE Bondad 6,100 ft., 1; Bondad, 15 mi. S Durango 6,050 ft., 1. *Archuleta Co.*: Chromo, 1 CM. *Las Animas Co.*: Trinidad, 6 BS. *Baca Co.*: unspecified, 1.

New Mexico: *San Juan Co.*: Blanco, 1 BS; Chuska Mountains, 8 BS. *Rio Arriba Co.*: 8 mi. N El Rito, 1; 4 mi. N El Rito, 5; Rim Rock, El Rito, 2; 2 mi. E El Rito, 7,000 ft., 1; 2 mi. SE El Rito, 1; 6 mi. E and ½ mi. S Truchas, 8,500 ft., 1; 2 mi. S and 4 mi. W Coyote, 8,100 ft., 1; unspecified, 2. *Taos Co.*: 3 mi. N Taos Pueblo, 5 BS; 23 mi. S and 6 mi. E Taos, 8,750 ft., 2. *Union Co.*: Emery Peak, 1 BS; Folsom, 3 BS; Sierra Grande, 8 BS; unspecified, 2. *McKinley Co.*: Bear Ridge, Zuni Mountains, 9 BS. *Sandoval Co.*: Bear Canyon, W foothills, Sandia Mountains, 3 BS; W foothills, near S end, Sandia Mountains, 7 BS. *Santa Fe Co.*: San Pedro, 7 BS. *San Miguel Co.*: Canadian River, 4 mi. NW Tucumcari, 1 BS. *Valencia Co.*: Mount Taylor, San Mateo Mountains, 10 BS.

Oklahoma: *Cimarron Co.*: Kenton, 1 BS.

***Eutamias quadrivittatus hopiensis* Merriam**

Eutamias hopiensis Merriam, Proc. Biol. Soc. Washington, 18:165, June 29, 1905.

Eutamias quadrivittatus hopiensis, Howell, Jour. Mamm. 3:184, August 4, 1922.

Type.—Female, adult, skull and skin, No. 67768 U. S. Nat. Mus.; from Keams Canyon, Painted Desert, Arizona; obtained on July 27, 1894, by A. K. Fisher.

Diagnosis.—Size medium; dorsal dark stripes tawny; crown Drab-Gray; baculum of same proportions as in *E. q. quadrivittatus* but smaller.

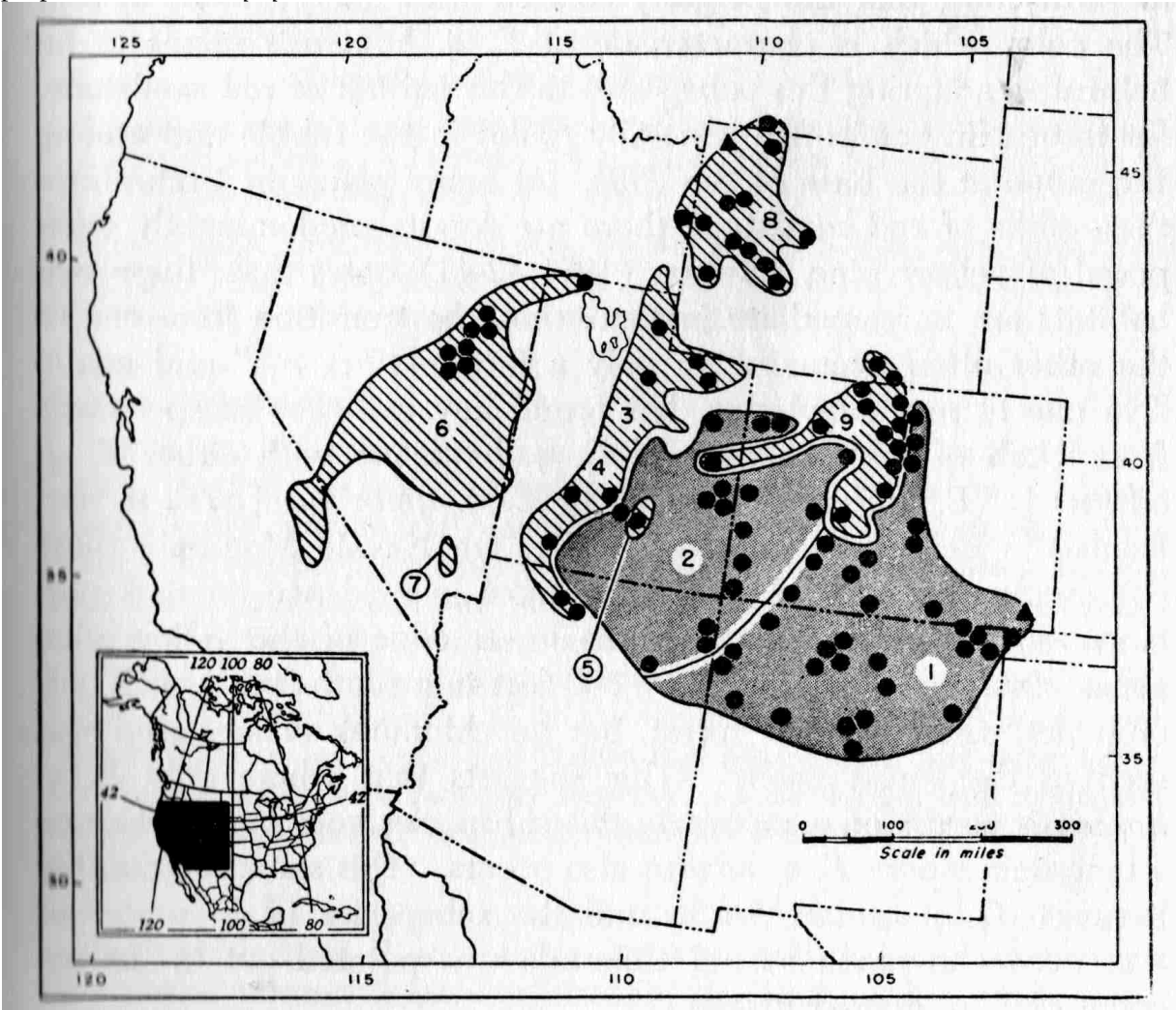


FIG. 2. Localities of specimens examined and probable geographic ranges of the subspecies of *Eutamias quadrivittatus* and *Eutamias umbrinus*. The symbols for locality records are as follows: circles, precise localities; triangles, localities known only to county.

Guide to subspecies:

1. *E. q. quadrivittatus*
2. *E. q. hopiensis*
3. *E. u. umbrinus*

4. *E. u. adsitus*
5. *E. u. sedulus*
6. *E. u. inyoensis*
7. *E. u. nevadensis*
8. *E. u. fremonti*
9. *E. u. montanus*

Description.—*Color pattern:* Head Drab-Gray, with Snuff Brown around margin of crown; facial stripes Sayal Brown with small blackish patches around eye; ears Ochraceous Tawny anteriorly and Pinkish Buff posteriorly; dorsal stripes Tawny, median one sometimes blackish; median pair of dorsal light stripes grayish white, outer pair creamy white; sides Ochraceous Tawny; rump and thighs Cinnamon Buff washed with Pale Smoke Gray; antipalmar and antiplantar surfaces of feet Pinkish Cinnamon; dorsal surface of tail Fuscous Black; ventral surface of tail Ochraceous Tawny, Fuscous Black along margin, Cinnamon Buff along outermost edge; underparts creamy white. *Skull:* As in *E. q. quadrivittatus*. *Baculum:* Same proportions as in *E. q. quadrivittatus* but smaller.

Comparisons.—See under the account of *E. q. quadrivittatus*.

Remarks.—Topotypes of this subspecies are intergrades between it and *E. q. quadrivittatus*.

In a large part of the geographic range of *E. q. hopiensis* there are numerous, massive outcrops of Mesozoic sandstones, which tend to form cliffs, that are brightly colored with many shades of red. The color which is characteristic of *E. q. hopiensis* seems to be helpful in adapting this subspecies to this habitat of red sandstone, for these chipmunks are generally found in the rubble and among the piñon at the base of the cliffs. At many places in Utah above these cliffs of red sandstone there are forests predominantly composed of yellow pine. Kelson (1951:42-43) states that "these two habitats are in immediate juxtaposition, the transition from one to the other often occurring in only a few feet ..." and again, "No one to my knowledge, has found any evidence in specimens from Utah of interbreeding of *E. q. hopiensis* with either *E. q. adsitus* [= *E. umbrinus adsitus*] or *E. q. umbrinus* [= *E. u. umbrinus*]." Benson (1935:449) states, "On Navajo Mountain these chipmunks [*E. q. hopiensis*] were most in evidence on rock outcrops surrounded by brush at the lower edge of the yellow pine zone. One was seen at about 9,500 feet in a south-facing rock outcrop near the spruce-fir forest, but no chipmunk of any kind was seen in the forest itself." This suggests that where only *E. q. hopiensis* occurs on a mountain this subspecies goes higher than on a mountain where *E. u. adsitus* also occurs. This same relationship between *E. q. quadrivittatus* and

the subspecies of *E. umbrinus* that occurs in north-central Colorado was pointed out in the account of *E. q. quadrivittatus*.

Specimens examined.—Total number, 68.

Utah: *Uintah Co.*: E side of confluence of Green and White rivers, 1 mi. SE Ouray, 4,700 ft., 3 UU. *Grand Co.*: Colorado River above Moab, 1 UU; side canyon of Colorado River above Moab, 1 UU; Moab, up Colorado River, 1 UU; Moab, 4,500 ft., 4 UU; Moab Bridge over Colorado River, 3,995 ft., Moab, 1 UU; Colorado River, 5 mi. E Moab Bridge, 4,000 ft., 1 UU. *Wayne Co.*: Fruita, 1 UU.

Colorado: *Moffat Co.*: 11 mi. W and 11 mi. N Rangely, 6,000 ft., 3. *Rio Blanco Co.*: White River, 5 BS. *Eagle Co.*: McCoy, 2 BS. *Mesa Co.*: 1½ mi. S Loma, 4,600 ft., 1. *Gunnison Co.*: 1 mi. E Somerset, 6,100 ft., 1. *Montrose Co.*: 1 mi. E Naturita, 5,900 ft., 1. *Dolores Co.*: 1 mi. N Cahone, 6,900 ft., 1. *Montezuma Co.*: 1 mi. S Cortez, 5,000 ft., 1; Mesa Verde, 25 mi. SW Mancos, 7,000 ft., 2 BS.

Arizona: *Navajo Co.*: Keams Canyon, 80 mi. N Holbrook, 15 BS. *Apache Co.*: Summit, 8,000 ft., Luka Chukai Mountains, 15 mi. E Luka Chukai Navajo School, 8 BS; Wheatfield Creek, W slope Tunicha Mountains, 7,000 ft., 3 BS.

***Eutamias umbrinus* (J. A. Allen)**

Diagnosis.—Size medium; pelage dark; sides dark; narrow cranial breadth; baculum distinguishable from that of any other species (*E. palmeri* excepted) by the combination of width of base more than ⅓ of length of shaft, distal ½ of shaft laterally compressed, and keel ¼ of length of tip.

***Eutamias umbrinus umbrinus* (J. A. Allen)**

Tamias umbrinus J. A. Allen, Bull. Amer. Mus. Nat. Hist. 3:96, June, 1890.

Eutamias umbrinus, Miller and Rehn, Proc. Boston Soc. Nat. Hist. 30:45, December 27, 1901.

Type.—Male, adult, skull and skin, No. 186463 U. S. Nat. Mus., Biol. Surv. Coll.; from Blacks Fork, about 9,500 ft., Uinta Mountains, Utah; obtained on September 19, 1888, by Vernon Bailey; original No. 228.

Diagnosis.—Size medium; general tone of upper parts dark and shadowy; skull relatively small.

Description.—*Color pattern*: Head Pale Smoke Gray; facial stripes Fuscous Black to Snuff Brown; ear Fuscous Black; posterior margin of ear and postauricular patch grayish white; median dorsal dark stripe black with Sayal Brown along margins; lateral pair of dorsal dark stripes Sayal Brown or Fuscous Black mixed with Sayal Brown; outermost pair of dorsal dark stripes nearly absent; sides Sayal Brown mixed with Cinnamon; rump and thighs Sayal Brown mixed with Smoke Gray; antipalmar and antiplantar surfaces of feet Cinnamon-Beige; ventral surface of tail Ochraceous Tawny or Sayal Brown, with Fuscous Black around margin and Pinkish Beige around outermost edge; underparts creamy white with dark gray underfur. *Skull*: Large, with moderately inflated braincase and well developed zygomatics. *Baculum*: One of the largest in the species.

Comparisons.—From *Eutamias umbrinus adsitus*, the subspecies to the south on the Wasatch Range, *E. u. umbrinus* differs in: Sides lighter; rump browner; hairs around outermost edge of tail tawnier (in freshly molted tails); shorter inner mandibular length.

From *E. u. inyoensis*, the subspecies to the west in central and northeastern Nevada and in northwestern Utah, *E. u. umbrinus* differs in: General tone of upper parts lighter; sides lighter; total length more; interorbital region broader.

For comparisons with *E. u. sedulus*, *E. u. fremonti*, and *E. u. montanus*, see the accounts of those subspecies.

Specimens examined.—Total number, 55.

Wyoming: *Uinta Co.*: 9 mi. S Robertson, 8,000 ft., 15; 10 mi. S and 1 mi. W Robertson, 8,700 ft., 5; 11½ mi. S and 2 mi. E Robertson, 9,200 ft., 1; 2 mi. E and 12 mi. S Robertson, Ashley Nat. For., 1; 13 mi. S and 2 mi. E Robertson, 9,200 ft., 1.

Utah: *Rich Co.*: Monte Cristo, 18 mi. W Woodruff, 8,000 ft., 2 UU. *Summit Co.*: 13½ mi. S and 2 mi. E Robertson [Wyoming], 4; 1 mi. N Bridger Lake R. S., 9,400 ft., 4. *Wasatch Co.*: Snake Creek Canyon, 3 mi. NW Midway, 6,000 ft., 1 UU. *Uintah Co.*: Paradise Park, 21 mi. W and 15 mi. N Vernal, 10,050 ft., 20.

***Eutamias umbrinus adsitus* J. A. Allen**

Eutamias adsitus J. A. Allen, Brooklyn Institute Mus. Sci. Bull. 1:118, March 31, 1905.

Type.—Unsexed adult, skull and skin, No. 28728 Amer. Mus. Nat. Hist.; from Briggs Meadow, 10,000 ft., Beaver Mountains, Utah; obtained on August 20, 1904, by George P. Engelhardt.

Diagnosis.—Size medium; sides dark; general tone of upper parts dark; dorsal light and dark stripes strongly contrasting.

Description.—*Color pattern*: Head Cinnamon mixed with grayish white; stripe on margin of crown Verona-Brown or Bister; ocular stripe Fuscous Black mixed with Sayal Brown; submalar stripe Sayal Brown; ear Fuscous, Sayal Brown along anterior margin and Smoke Gray along posterior margin and on postauricular patch; median dorsal stripe black; lateral dorsal dark stripes Fuscous Black mixed with Russet; outermost dorsal dark stripes slightly darker or indistinguishable from sides in color; dorsal light stripes grayish white with Mikado-Brown along margins; outermost pair of dorsal light stripes nearly pure white; sides Russet mixed with Cinnamon or Ochraceous-Tawny; rump and thighs Smoke Gray mixed with Cinnamon-Buff, with a larger or smaller number of Fuscous Black hairs; antipalmar and antiplantar surfaces of feet Cinnamon-Buff; dorsal surface of tail black; ventral surface of tail Sayal Brown to Tawny; underparts white with dark underfur. *Skull and Baculum*: As in *E. u. umbrinus*.

Comparisons.—From *E. u. inyoensis*, the subspecies to the west, *E. u. adsitus* differs in: General tone of upper parts darker; sides darker; interorbital region wider; skull significantly deeper.

For comparison with *E. u. umbrinus*, *E. u. sedulus*, and *E. u. montanus*, see the accounts of those subspecies.

Remarks.—Specimens from West Rim, Zion National Park, 6,500 ft., Washington County, Utah, seem to be intergrades between *E. u. adsitus* and

E. u. inyoensis, and are referable to *E. u. adsitus*.

Specimens examined.—Total number, 34.

Utah: *Beaver Co.:* Britts Meadow, Beaver Range Mountains, 8,500 ft., 13 BS. *Wayne Co.:* Donkey Lake, Boulder Mountain, 10,000 ft., 4 UU. *Garfield Co.:* Wildcat R. S., Boulder Mountain, 8,700 ft., 5 UU.

Arizona: *Coconino Co.:* De Motte Park, Kaibab Plateau, 3 BS; Bright Angel, Kaibab Plateau, 9 BS.

***Eutamias umbrinus sedulus* new subspecies**

Type.—Male, adult, skull, skin, and baculum, No. 158181 U. S. Nat. Mus. Biol. Surv. Coll.; from Mount Ellen, Henry Mountains, Garfield County, Utah; obtained on October 13, 1908 by W. H. Osgood; original No. 3667.

Diagnosis.—Size medium; general tone of upper parts dark reddish-brown; ventral surface of tail Ochraceous-Orange; sides Mars Yellow.

Description.—*Color pattern:* Crown Drab-Gray mixed with Fuscous; upper facial stripe Fuscous Black mixed with Sudan Brown; ocular stripe Sudan Brown mixed with black; submalar stripe Sudan Brown slightly mixed with black; anterior margin of ear Sudan Brown slightly mixed with black; hairs inside pinna, posteriorly, Warm Buff; posterior margin of ear and postauricular patch creamy white; median dorsal dark stripe black with Antique Brown along margins; lateral dorsal dark stripes black mixed with Antique Brown; outermost dorsal dark stripes Xanthine Orange slightly mixed with black; median dorsal light stripes Pale Smoke Gray; outermost dorsal light stripes white slightly mixed with gray; rump and thighs Smoke Gray; sides Mars Yellow; dorsal surface of tail black mixed with Warm Buff; ventral surface of tail Ochraceous-Orange, with black around margin, and Warm Buff around outermost edge; antipalmar and antiplantar surfaces of feet Ochraceous-Buff; underparts creamy white with dark underfur. *Skull:* Large; braincase moderately inflated; zygomata strong. *Baculum:* As in *E. u. umbrinus*.

Comparisons.—From *E. u. umbrinus*, the subspecies from the Uinta and northern Wasatch Mountains of Utah, *E. u. sedulus* differs in: General tone of upper parts lighter; sides lighter.

From *E. u. adsitus*, the subspecies from the southern Wasatch Range in Utah and Kaibab Plateau in Arizona, *E. u. sedulus* differs in: Sides lighter; general tone of upper parts markedly lighter.

From *E. u. inyoensis*, the subspecies from central and northeastern Nevada, and western and northwestern Utah, *E. u. sedulus* differs in: Sides lighter (less grayish); general tone of upper parts tawnier.

For comparison with *E. u. montanus*, see the account of that subspecies.

Specimens examined.—Total number, 7 BS, all from the type locality.

***Eutamias umbrinus inyoensis* Merriam**

Eutamias speciosus inyoensis Merriam, Proc. Biol. Soc. Washington, 11:202, 208, July 1, 1897.

Type.—Male, adult, skull and skin, No. 29387/41462 U. S. Nat. Mus. Biol. Surv. Coll.; from Black Canyon, 8,200 ft., White Mountains, Inyo County, California; obtained on July 7, 1891, by E. W.

Nelson; original No. 1069.

Diagnosis.—Size medium; sides light; general tone of upper parts light; baculum one of largest in species.

Description.—*Color pattern*: Head Smoke Gray mixed with Pink-Cinnamon; upper two pairs of facial stripes Fuscous Black or black; submalar stripe Sayal Brown; ear Fuscous or Chaetura-Drab, posterior margin and postauricular patch buffy white; median dorsal dark stripe black with Sayal Brown along margins; lateral dorsal dark stripes black mixed with Sayal Brown or Mikado Brown; outermost dorsal dark stripes Sayal Brown or Mikado Brown mixed with black; sides Ochraceous-Tawny or Tawny; thighs Cinnamon-Buff mixed with Smoke Gray; antipalmar and antiplantar surfaces of feet Cinnamon-Buff; ventral surface of tail Cinnamon-Buff or Ochraceous-Tawny with Fuscous Black around margin and Pinkish Buff around outermost edge; underparts creamy white. *Skull*: Large; zygomata strong; braincase moderately inflated. *Baculum*: One of largest in species.

Comparisons.—For comparisons with *E. u. umbrinus*, *E. u. adsitus*, *E. u. sedulus*, and *E. u. nevadensis*, see the accounts of those subspecies.

Remarks.—The baculum in *E. u. inyoensis* is like that in *E. palmeri*.

Specimens examined.—Total number, 46.

Nevada: *Elko Co.*: Head Ackler Creek, N end Ruby Mountains, 1; Steels Creek, N end Ruby Mountains, 1; Summit Secret Pass, 6,200 ft., Ruby Mountains, 2; Three Lakes, Ruby Mountains, 11; Long Creek, S fork, Ruby Mountains, 4; Harrison Pass R. S., Green Mountain Canyon, 1; W side Ruby Lake, 6 mi. N Elko Co. line, 3; W side Ruby Lake, 3 mi. N Elko Co. line, 8. *White Pine Co.*: Willow Creek, 2 mi. S White Pine Co. line, Ruby Mountains, 6; W side Ruby Lake, 3 mi. S White Pine Co. line, 5; Overland Pass, E slope Ruby Mountains, 8 mi. S White Pine Co. line, 2.

Utah: *Boxelder Co.*: Head of George Creek and Clear Creek, 5 mi. S Stanrod, Raft River Mountains, 8,500 ft., 2 UU.

***Eutamias umbrinus nevadensis* Burt**

Eutamias quadrivittatus nevadensis Burt, Jour. Mamm. 12:299, August 24, 1931.

Type.—Male, adult, skull and skin, No. 15884 Donald R. Dickey Collection; from Hidden Forest, Sheep Mountains, 8,500 ft., Clark County, Nevada; obtained on July 13, 1929, by W. H. Burt; original No. 2337.

Diagnosis.—Size medium; general tone of upper parts grayish; baculum one of the largest of species.

Description.—"General tone of upperparts grayish; median dorsal stripe, extending from crown between ears to rump, black faintly bordered with 'verona brown'; lateral dark dorsal stripes similar to median stripe, but with anterior one-third deeply suffused with 'verona brown'; central light dorsal stripes grayish, slightly lighter than head and rump; lateral stripes white; head and rump 'pale smoke gray'; postauricular patch grayish white, a narrow margin extending up posterior border of ear; anterior portion of ear 'fuscous black' mixed with 'verona brown' at base and bordered by light gray; ocular stripe black grading into 'verona brown' in front of ear; submalar stripe nearly obsolete, 'sayal brown'; sides of body grayish washed with 'verona brown'; feet grayish very faintly washed with 'pinkish buff'; dorsal surface of tail black overlaid with 'tilleul buff'; ventral surface of tail 'cinnamon buff' narrowly bordered by black then by 'tilleul buff'; ventral surface of body white." (Burt 1931:299.) Skull similar to that of *E. u. inyoensis* but differing as indicated below.

Comparisons.—From *E. u. inyoensis*, the subspecies to the north, *E. u. nevadensis* differs in: Paler and grayer throughout; tawny areas restricted; gray areas clearer and less suffused; dark facial markings narrower and less distinct; ventral surface of tail distinctly paler; feet lighter, clearer gray; nasals extend farther posteriorly with respect to premaxillae (Burt *loc. cit.*).

From *E. u. adsitus*, the subspecies to the northeast, *E. u. nevadensis* differs in: Narrower dorsal stripes and facial markings; paler coloration of head, rump, sides, feet, and ventral surface of tail (Burt *op. cit.*: 299-300).

Remarks.—The differences between *E. umbrinus nevadensis* and *E. palmeri*, as shown by Burt (*op. cit.*) and Hall (1946), are such that one might expect *E. palmeri* to be a subspecies of *E. umbrinus*. However, having only the structure of the baculum as evidence additional to that summarized by Hall (*op. cit.*), I follow him in according *E. palmeri* specific status.

Specimens examined.—None.

***Eutamias umbrinus fremonti* new subspecies**

Type.—Male, adult, skull, skin, and baculum, No. 41790 Univ. Kansas Mus. Nat. Hist.; from 31 mi. N Pinedale, 8,025 ft., Sublette County, Wyoming; obtained on July 8, 1951, by Rollin H. Baker; original No. 1596.

Diagnosis.—Size large; sides Capucine Yellow; antiplantar surface of hind feet Raw Sienna; postauricular patch grayish white; baculum as in *E. u. umbrinus*.

Description.—*Color pattern*: Crown Cinnamon-Buff mixed with gray; upper facial stripe Sepia; ocular stripe Chaetura-Drab; submalar stripe Fuscous Black mixed with Sayal Brown; ear black; anterior margin of ear Mars-Yellow, posterior margin grayish white; hairs inside posterior portion of pinna Dresden-Brown; postauricular patch Pale Smoke Gray; median dorsal dark stripe black; lateral dorsal dark stripe black mixed with Sayal Brown; outermost dorsal dark stripe obsolete, Buckhorn-Brown mixed with black; median pair of dorsal light stripes grayish mixed with Buckhorn-Brown; outer pair of dorsal light stripes creamy white; sides Buckhorn-Brown; rump Pale Smoke Gray mixed with Saccardo's Umber; dorsal surface of tail black mixed with Buckhorn-Brown; ventral surface of tail Sayal Brown; outermost edge of tail Light Buff; antipalmar surface of forefeet Warm Buff; antiplantar surface of hind foot Ochraceous-Tawny; underparts creamy white with dark underfur. *Skull*: Large, with strong zygomata; braincase well inflated. *Baculum*: As in *E. u. umbrinus*.

Comparisons.—From *E. u. umbrinus*, the subspecies from the Uinta and northern Wasatch Mountains in Utah, *E. u. fremonti* differs in: Sides darker; antiplantar surfaces of feet darker; postauricular patch grayer; crown more grayish; skull slightly larger.

From *E. ruficaudus ruficaudus*, the species and subspecies from western Montana, *E. u. fremonti* differs in: General tone of upper parts, sides, underside of tail, and feet, all darker in coloration; baculum shorter and proportionally twice as wide at base.

For comparison with *E. u. montanus*, see the account of that subspecies.

Remarks.—The geographic ranges of *E. umbrinus fremonti* and *E. ruficaudus ruficaudus* are allopatric and no specimens have ever been taken in the intermediate area to indicate whether or not these two species anywhere occur together. The bacula in the two species differ to the same degree as those of *E. quadrivittatus* and *E. umbrinus*. The differences between *E. u. fremonti* and *E. r. ruficaudus* are such that in my opinion, *E. ruficaudus* is a distinct species.

Specimens examined.—Total number, 58.

Montana: Park Co.: Beartooth Mountains, 2 BS.

Idaho: Bonneville Co.: Big Hole Mountains, 9,000 ft., near Irwin, 1 BS.

Wyoming: Yellowstone Park, 2. Park Co.: 16¼ mi. N and 17 mi. W Cody, 5,625 ft., 2. Teton Co.: 1 mi. E and ¼ mi. N Togwotee Pass, 9,800 ft., 2; Amphitheatre Lake, Teton Park, 1 MM; Flat Creek, 4 MM; head of Cache Creek, 4 MM; Jackson, Upper Arizona Creek, 2 MM; Flat Creek-Granite Creek divide, 6 MM; Flat Creek Pass, 1 MM; Flat Creek-Gravel Creek divide, 2 MM. Lincoln Co.: La Barge Creek, 9,000 ft., 2 BS. Fremont Co.: Togwotee Pass, 12; 12 mi. N and 3 mi. W Shoshoni, 4,650 ft., 1; Mosquito Park R. S., 9,500 ft., 17½ mi. W and 2½ mi. N Lander, 1; 17 mi. S and 6½ mi. W Lander, 8,450 ft., 3. Sublette Co.: 31 mi. N Pinedale, 8,025 ft., 2; W side Barbara Lake, 10,300 ft., 8 mi. S and 3 mi. W Fremont Peak, 4; 19 mi. W and 2 mi. S Big Piney, 7,700 ft., 5.

***Eutamias umbrinus montanus* new subspecies**

Type.—Male, adult, skull, skin, and baculum, No. 20105 Univ. Kansas Mus. Nat. Hist.; from ½ mi. E and 3 mi. S Ward, 9,400 ft., Boulder County, Colorado; obtained on August 1, 1947, by E. L. Cockrum; original No. 721.

Diagnosis.—Size large; sides Clay Color; antipalmar and antiplantar surfaces of feet Cinnamon-Beff; baculum as in *E. u. umbrinus*.

Description.—*Color pattern:* Crown Raw Sienna mixed with gray; upper facial stripe and ocular stripe black mixed with Sepia; submalar stripe Snuff Brown mixed with black; ear black or Sepia, anterior margin Ochraceous-Tawny, posterior margin and postauricular patch grayish white; hairs inside posterior part of pinna Cinnamon-Beff; median dorsal dark stripe black with Sayal Brown along margins; lateral dark stripes black mixed with Sayal Brown; outermost dorsal dark stripes obsolete, Sayal Brown mixed with black; median pair of dorsal light stripes Pale Smoke Gray mixed with Clay Color; outer pair of dorsal light stripes creamy white; sides Clay Color; rump and thighs Neutral Gray; dorsal surface of tail black mixed with Cinnamon-Beff; ventral surface of tail Ochraceous-Tawny; hairs around margin of tail Cinnamon-Beff or Ochraceous-Tawny; antipalmar and antiplantar surfaces of feet Cinnamon-Beff; underparts creamy white with dark underfur. *Skull:* Large; zygomata strong; braincase well inflated. *Baculum:* As in *E. u. umbrinus*.

Comparisons.—From *E. quadrivittatus quadrivittatus*, the subspecies and species to the south, *E. u. montanus* differs in: General tone of upper parts darker; braincase significantly narrower; baculum shorter and markedly wider at base.

From *E. u. umbrinus*, the subspecies from the Uinta and northern Wasatch Mountains, *E. u. montanus* differs in: General tone of upper parts brighter (less tawny); sides more tawny; skull slightly larger.

From *E. u. sedulus*, the subspecies from the Henry Mountains of Utah, *E. u. montanus* differs in: Sides darker; general tone of upper parts darker.

From *E. u. fremonti*, the subspecies from the mountains of western and northwestern Wyoming, *E. u. montanus* differs in: General tone of upper parts lighter; hairs around outermost edge of tail tawnier.

Remarks.—Howell (1929:83) stated that the specimens of *E. quadrivittatus quadrivittatus* (= *E. umbrinus montanus*) from Estes Park, Long's Peak, and Gold Hill, all in Colorado, "average somewhat darker on the back and sides than typical *quadrivittatus*; the light dorsal stripes are also somewhat duller and the dark stripes less blackish, thus showing an approach to the characters of *umbrinus*." Now there are more specimens of *E. u. montanus* from the mountains of north-central Colorado than were available to Howell. He was not aware of the striking difference between the bacula of *E. quadrivittatus* and *E. umbrinus*, and the constancy of this difference between all the subspecies of one species and those of the other.

Although the geographic range of *E. u. umbrinus* is closer to the ranges of *E. u. fremonti* and *E. u. montanus* than to the geographic range of *E. u. adsitus*, *E. u. umbrinus* seems to be more closely related to *E. u. adsitus* than to *E. u. fremonti* or *E. u. montanus*. This observation may be explained by the presence of continuous habitat for *E. umbrinus* between the ranges of *E. u. umbrinus* and *E. u. adsitus*, whereas *E. u. fremonti* and *E. u. montanus* are each separated from *E. u. umbrinus* by areas unsuitable for occupancy by *E. umbrinus*. It must be noted, however, that no actual intergrades between *E. u. umbrinus* and *E. u. adsitus* are known.

Specimens examined.—Total number, 36.

Wyoming: Albany Co.: 3 mi. ESE Brown's Peak, 10,000 ft., 2; 3½ mi. S Wood's Landing, 1.

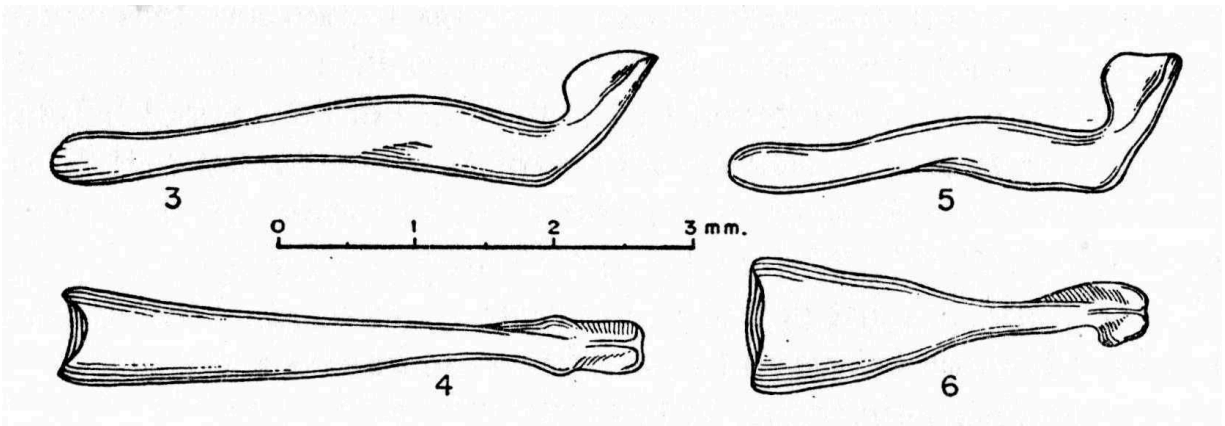
Utah: Uintah Co.: PR Springs, 7,950 ft., 43 mi. S Ouray, Uintah-Grand county line, 1 UU.

Colorado: Jackson Co.: Mount Zirkel, 10,000 ft., on trail, 2 WC; Buffalo Pass, 10,380 ft., 1 WC; Buffalo Pass road, 10,130 ft., 1 WC. Larimer Co.: 2 mi. E Log Cabin, 7,450 ft., 1 WC; Estes Park, 7,600 ft., 1; 1½ mi. SW Estes Park, 1; 2½ mi. SW Estes Park, 2; 3½ mi. SW Estes Park, 1; 12 mi. SW Estes Park, 1. Rio Blanco Co.: 1 mi. NW Pagoda Peak, 10,400 ft., 1. Boulder Co.: Long's Peak, 7 BS; 1 mi. NE Ward, 10,000 ft., 1; 3 mi. S Ward, 9,000 ft., 5; ½ mi. E and 3 mi. S Ward, 9,400 ft., 1; 1 mi. S Gold Hill, 8,200 ft., 1. Clear Creek Co.: Davidson Mine, 3 mi. SW Idaho Springs, 1; Georgetown, 1 CM. Jefferson Co.: Silver Plume, 1 CM. Park Co.: Tarryall Creek Camp, 8,700 ft., 1 WC. Gunnison Co.: S side Crested Butte Mountain, 9,500 ft., 1 WC; mouth of Virginia Basin, Gothic, 1 FC.

DISCUSSION

The chipmunks that heretofore have been assigned to the species *Eutamias quadrivittatus* are here assigned to two species, *E. quadrivittatus* and *E. umbrinus*, for the following reasons:

1. The baculum of *E. quadrivittatus* differs from that of *E. umbrinus* in having a narrow base (see figs. 3, 4). This difference permits any specimen which has an associated baculum to be readily identified to species.
2. The cranial breadth in the subspecies of *E. quadrivittatus* is significantly larger than in the subspecies of *E. umbrinus*.
3. Specimens of *E. umbrinus* are darker than any specimen of *E. quadrivittatus*.
4. Where the geographic ranges of *E. quadrivittatus* and *E. umbrinus* come close to one another (probably they meet at some places), *E. umbrinus* occupies a higher position in terms of life-zones. Wherever either of these two species, but not the other, occurs on a mountain the species occupies both the higher and lower life-zones.



FIGS. 3 and 4. Baculum of *Eutamias quadrivittatus quadrivittatus*, No. 35648/47919 BS; from Canon City, Fremont Co., Colorado. Figure 3, lateral view of right side; figure 4, dorsal view.

FIG. 5. Baculum of *Eutamias umbrinus umbrinus*, No. 38062; from Paradise Park, 21 mi. W and 15 mi. N Vernal, 10,050 ft., Uintah Co., Utah. Lateral view of right side.

FIG. 6. Baculum of *Eutamias umbrinus montanus*, No. 20105; from ½ mi. E and 3 mi. S Ward, 9,400 ft., Boulder Co., Colorado. Dorsal view.

The differences between *E. quadrivittatus* and *E. umbrinus* are as great as, or greater than, between many species of chipmunks, such as between *E. minimus* and *E. amoenus*, and between *E. quadrivittatus* and *E. cinereicollis*.

Although I know of no ecological differences between *E. umbrinus* and *E. ruficaudus*, the morphological differences, as for example, differences in the structure of the baculum, and differences in color pattern, lead me to maintain *E. ruficaudus* and *E. umbrinus* as separate species.

The present distribution of these two species is attributable to the uplift of the Rocky Mountains in the Pleistocene. That the uplift of the Rocky Mountains and the erosion which produced the present-day relief took place in Pleistocene times is supported by the evidence found by several geologists such as Hunt and Sokoloff (1950:109-123).

The present geographic distribution of *E. umbrinus* and *E. quadrivittatus* conceivably came about as follows: *E. umbrinus*-like chipmunks were present, before the uplift of the major chains of mountains, on isolated, low mountain ranges that were not covered with glaciers (such as the laccolithic mountains that occur in Utah) in Pleistocene time, while *E. quadrivittatus*-like chipmunks were present in the central parts of New Mexico, Colorado, and southern Wyoming. With the advent of uplift, the habitats in the central parts of these states were changed from a plains-like habitat to a habitat that resembled the forest habitats that exist today. *E. umbrinus*-like chipmunks then invaded this newly formed habitat and displaced any *E. quadrivittatus*-like chipmunks that were less well adapted to live there. The Colorado River probably served as a barrier that kept the *E. umbrinus*-like chipmunks and *E. quadrivittatus*-like chipmunks separated up to this time. Invasion of the new forest-niche by *E. umbrinus*-like chipmunks may have taken place through the Uinta Mountains of northeastern Utah, after the glaciers disappeared from these mountains, since the Colorado River probably prevented any eastward migration farther south.

TABLE 1

**Average and Extreme Measurements in Millimeters of Adult
Eutamias quadrivittatus and *E. umbrinus***

	Greatest length of skull	Zygomatic breadth	Cranial breadth	Length of nasals	Total length	Length of tail	Length of lower tooth- row	Condyl- alveolar length of mandible
<i>E. q. quadrivittatus</i> , Canon City, Fremont Co., Colorado.								
Mean (7)	35.7	19.3	16.2	11.0	222	99.4	5.40	18.98
Min ♂	35.7	19.1	15.9	10.6	216	93.0	5.32	18.65
Max	35.8	19.8	16.5	11.6	230	104.0	5.49	19.41
Mean (3)	35.9	19.9	16.5	10.9	231	99.0	5.42	19.10
Min ♀	35.6	19.9	16.5	10.7	200	98.0	5.39	18.85
Max	36.2	19.9	16.6	11.2	232	100.0	5.49	19.28
<i>E. q. hopiensis</i> , Moab, Grand Co., Utah.								
Mean (11)	34.4	19.4	16.3	10.0	212	90.7	5.19	18.36
Min ♂	33.5	19.2	15.9	9.3	208	85.0	4.92	17.80
Max	35.4	20.0	16.8	10.5	220	96.0	5.38	18.96
Mean (5)	34.9	19.6	16.4	10.5	219	94.4	5.16	18.58
Min ♀	34.2	19.3	15.9	10.1	210	85.0	5.13	18.00
Max	35.7	20.1	16.6	10.7	228	104.0	5.20	19.19
<i>E. u. umbrinus</i> , Mts. S Robertson, Uintah Co., Wyoming.								
Mean (11)	34.7	18.9	15.7	10.9	218	96.2	5.13	18.04
Min ♂	34.3	18.3	15.6	10.3	215	81.0	4.79	17.57
Max	35.2	19.4	16.0	11.7	228	112.0	5.42	18.59
	35.1	19.2	15.9	11.0	224	96.4	5.17	18.46
Mean (4)	34.9	18.6	15.7	10.3	204	90.0	5.11	18.31
	35.4	20.0	16.2	11.8	234	100.0	5.22	18.98

Min ♀
Max

E. u. adsitus, Britts Meadow, Beaver Co., Utah.

Mean (6)	34.8	18.9	15.5	10.8	214	89.6	5.16	18.07
Min ♂	34.3	18.5	15.3	10.4	203	73.0	4.64	17.69
Max	35.4	19.6	16.1	11.3	225	95.0	5.34	18.70

Mean (6)	35.1	19.5	16.0	11.0	228	96.5	5.11	18.75
Min ♀	33.9	18.9	15.9	10.6	215	95.0	5.00	18.51
Max	36.2	20.0	16.3	11.8	233	98.0	5.33	19.40

E. u. sedulus, Mt. Ellen, Henry Mts., Garfield Co., Utah.

Mean (5)	34.7	18.7	15.6	10.7	218	93.0	5.21	18.74
Min ♂	33.5	18.4	15.4	10.1	213	89.0	5.09	18.48
Max	35.5	19.1	15.9	11.2	224	97.0	5.28	19.38

Mean (2)	34.9	19.4	16.1	11.1	227	98.0	5.24	18.74
Min ♀	34.9	19.3	16.1	11.0	224	96.0	5.07	19.24
Max	34.9	19.5	16.1	11.3	231	100.0	5.42	19.80

E. u. inyoensis, Ruby Mts., Elko and White Pine Cos., Nev.

Mean (12)	34.4	19.0	15.7	10.5	208	89.5	5.15	18.12
Min ♂	33.5	18.6	15.2	10.0	196	85.0	5.01	17.32
Max	35.4	19.6	16.1	11.5	220	100.0	5.37	18.81

Mean (5)	34.9	19.4	15.7	10.5	215	92.8	5.19	18.63
Min ♀	34.4	19.1	15.4	10.4	204	86.0	5.04	18.50
Max	35.2	19.7	16.0	10.7	226	102.0	5.33	18.80

E. u. nevadensis, Measurements of the type (Burt 1931:300).

♂	34.8	19.3	16.2	11.0	205	89.0	...	...
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E. u. fremonti, Togwotee Pass, Fremont Co., Wyoming.

Mean (8)	35.6	19.3	15.9	11.4	223	99.0	5.34	19.17
Min ♂	35.2	18.9	15.8	11.1	216	95.0	5.22	18.72
Max	36.5	19.7	16.1	11.8	243	111.0	5.57	19.78

Mean (6)	35.3	19.6	15.9	11.3	229	101.0	5.40	19.02
Min ♀	34.5	19.3	15.7	10.9	223	92.0	5.35	18.37
Max	36.0	20.0	16.5	12.0	239	110.0	5.44	19.51

E. u. montanus, Boulder Co., Colorado.

Mean (5)	35.2	18.8	15.5	10.8	226	96.0	5.20	18.29
Min ♂	34.7	18.4	15.2	10.1	215	93.0	5.03	17.80
Max	36.8	19.4	16.2	11.5	232	115.0	5.53	19.36

Mean (6)	35.7	19.1	15.6	10.9	226	98.0	5.28	18.67
Min ♀	35.1	18.8	15.1	10.3	215	89.0	5.06	18.09
Max	36.5	19.5	16.0	11.6	231	105.0	5.58	19.35

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